

Infantile-Onset Pompe Disease (IOPD): A Comprehensive Disease-Characterization Report

A Mendelian lysosomal storage disorder — glycogen storage disease type II, infantile form

Summary

Infantile-Onset Pompe Disease (IOPD) is a severe, autosomal-recessive lysosomal storage disorder (glycogen storage disease type II; OMIM #232300; MONDO:0009290; Orphanet ORPHA:79237) caused by biallelic loss-of-function variants in the *GAA* gene (17q25.3; HGNC:4065), which encodes lysosomal acid α -glucosidase (acid maltase; EC 3.2.1.20; UniProt P10253). Complete or near-complete loss of *GAA* enzyme activity prevents the lysosomal breakdown of glycogen to glucose, driving massive intralysosomal glycogen accumulation in cardiac, skeletal, and smooth muscle, and — as ERT-prolonged survival has unmasked — throughout the central nervous system. Untreated classic IOPD presents in the first months of life with a "floppy infant" picture of profound generalized hypotonia, rapidly progressive hypertrophic cardiomyopathy (with a pathognomonic short-PR/high-voltage-QRS/Wolff-Parkinson-White ECG signature), feeding and respiratory difficulty, and is uniformly fatal — usually by 1–2 years of age from cardiorespiratory failure.

The disease has been transformed by **enzyme replacement therapy (ERT)**. First-generation recombinant human *GAA* (alglucosidase alfa, FDA-approved 2006) prolongs ventilator-free survival and reverses cardiomyopathy, but its benefit is critically modulated by **CRIM (cross-reactive immunological material) status** and by the **age at which treatment begins**. Approximately 85–87% of IOPD patients are CRIM-positive; CRIM-negative patients cannot make any native *GAA*, mount high sustained anti-rh*GAA* antibody titers, and fail treatment unless given immune tolerance induction (rituximab + methotrexate $\pm$ IVIG). Newborn screening now enables ERT initiation in the neonatal window (≤ 4 weeks), improving outcomes, though residual skeletal-muscle disease, CNS involvement, and antibody risk persist. Next-generation ERTs — the bis-mannose-6-phosphate-enhanced **avalglucosidase alfa** and the second-generation **cipaglucosidase alfa plus the oral stabilizer miglustat** — address

intrinsic limitations of first-generation rhGAA (blood-pH inactivation, inefficient uptake, incomplete processing). Investigational AAV gene therapy corrects multi-tissue (including CNS) storage preclinically but carries dorsal-root-ganglion toxicity risk in humans.

This report synthesizes **16 evidence-backed findings** drawn from **62 reviewed papers** spanning human clinical cohorts, pivotal ERT trials, natural-history and genotype–phenotype studies, animal models, and mechanistic work. The mechanistic through-line: *GAA null variants → absent enzyme → lysosomal glycogen entrapment → glycogenosome rupture → secondary autophagic/mTORC1–AMPK dysregulation and a metabolic shift to lipid fuel → multisystem (cardiac, skeletal, respiratory, CNS) failure.*

Disease Overview and Identifiers (Section 1)

IOPD is the most severe end of the Pompe disease clinical spectrum. Pompe disease as a whole is defined by deficiency of the lysosomal enzyme acid α -glucosidase and consequent intralysosomal glycogen accumulation. As stated in a 2025 review, "*Pompe disease (PD) is a rare genetic disorder that leads to intralysosomal glycogen accumulation because of a deficiency in the lysosomal enzyme acid α -glucosidase (GAA), which is required to break down glycogen to glucose*" (PMID: 40237692). The infantile form is distinguished clinically by "*severe and progressive hypertrophic cardiomyopathy and muscle weakness with death in the first 2 years of life if left untreated*" (PMID: 39314994).

Resource	Identifier
OMIM	#232300 (glycogen storage disease II)
MONDO	MONDO:0009290 (glycogen storage disease II); IOPD subtype under Pompe disease
Orphanet	ORPHA:79237 (infantile-onset) / ORPHA:365 (Pompe)
ICD-10	E74.02 (Pompe disease)
ICD-11	5C51.3
MeSH	D006009 (Glycogen Storage Disease Type II)
Gene	GAA, HGNC:4065, NCBI Gene 2548, Ensembl ENSG00000171298
Protein	UniProt P10253 (lysosomal α -glucosidase)
Enzyme	EC 3.2.1.20

Synonyms / alternative names: Pompe disease (infantile form), glycogen storage disease type II (GSD II / GSDII), acid maltase deficiency, acid α -1,4-glucosidase deficiency, classic infantile Pompe disease, cardiomegalia glycogenica.

Source of information: The evidence base is largely aggregated disease-level (natural history studies, registries, clinical trials, systematic reviews) supplemented by individual case reports and single-center cohorts; it is not primarily EHR-derived.

Etiology (Section 2)

Primary cause — genetic. IOPD is a monogenic, autosomal-recessive disorder caused by **biallelic loss-of-function variants in GAA**. The disorder is "*caused by a deficiency in acid α -glucosidase (GAA), ... a lysosomal storage disorder*" (PMID: 41742217). There is no environmental or infectious cause.

Genetic risk factors. The disease-causal factors are the two inherited pathogenic GAA alleles. IOPD specifically results from **two "severe/null" variants** (nonsense, frameshift, large deletions, canonical splice) producing near-absent enzyme, in contrast to late-onset Pompe

disease (LOPD), where the common "leaky" splice variant **c.-32-13T>G (IVS1)** retains residual activity ([PMID: 38958145](#); [PMID: 34020684](#)). Genotype determines both phenotype severity and CRIM status.

Environmental / lifestyle / infectious factors. None are causal. IOPD is fully penetrant given a severe biallelic genotype; sex is not a risk factor (autosomal). Family history and consanguinity increase recurrence risk in the classic Mendelian sense.

Protective factors. No environmental protective factors are established. The principal disease-modifying factor is therapeutic (early ERT + immune tolerance). At the molecular level, any residual GAA activity (as conferred by "leaky" alleles) shifts phenotype toward the milder late-onset spectrum, functioning as an intrinsic genetic modifier of severity.

Gene–environment interactions. Not applicable in the classical toxicological sense. The clinically important "interaction" is between the patient's GAA genotype (CRIM status) and the therapeutic environment (ERT + immunomodulation): *"Variant pathogenicity correlated with both CRIM status and clinical outcomes"* ([PMID: 40952111](#)).

Phenotypes (Section 3)

Classic IOPD presents in the first months of life. In a case series of six Chinese infantile GSD II patients (onset birth to 3 months), *"All patients had varying degrees of generalized muscle weakness, hypotonia and development retardation or retrogression. Other common findings were feeding difficulties in two patients, tongue weakness in two patients, respiratory distress in four patients, macroglossia in one patient, and hepatomegaly in two patients. Left ventricular hypertrophy and cardiomegaly were obvious in all the six patients"* ([PMID: 26310554](#)).

Phenotype	Type	Frequency	Onset / progression	HPO term
Hypertrophic cardiomyopathy	Physical/ imaging	~100% (universal)	Neonatal– infantile, rapidly progressive	HP:0001639
Cardiomegaly	Imaging sign	~100%	Infantile, progressive	HP:0001640
Generalized muscle hypotonia ("floppy infant")	Clinical sign	~100%	Infantile, progressive	HP:0001290 / HP:0001252
Muscle weakness	Symptom	~100%	Infantile, progressive	HP:0001324
Motor delay / developmental regression	Sign	High	Infantile, progressive	HP:0001270
Respiratory distress / insufficiency	Sign	~65% (4/6)	Infantile, progressive	HP:0002093
Feeding difficulties	Symptom	~30%	Infantile	HP:0011968
Macroglossia	Physical	Variable (~17%)	Infantile	HP:0000158
Tongue weakness	Sign	~30%	Infantile	HP:0000191
Hepatomegaly	Sign	~30%	Infantile	HP:0002240
Short PR interval / WPW pre-excitation	Lab/ECG	Common	Infantile, may present	HP:0005165 / HP:0001728
Elevated creatine kinase	Lab abnormality	High	Infantile	HP:0003236
Sensorineural hearing loss (ERT survivors)	Sign	6/8 in long- term cohort	Emerges with prolonged survival	HP:0000407
Flaccid dysarthria (ERT survivors)	Sign	8/8		HP:0001260

Phenotype	Type	Frequency	Onset / progression	HPO term
			Emerges with prolonged survival	
Ptosis (ERT survivors)	Sign	~4/8	Later	HP:0000508

Severity and progression. Untreated classic IOPD is uniformly severe and relentlessly progressive, fatal within ~1–2 years. **Quality-of-life impact** is profound: infants lose feeding, motor, and respiratory function; long-term ERT survivors, though alive into adulthood, retain substantial multisystem morbidity — *"Two participants were ambulatory; 6 used wheelchairs. Flaccid dysarthria (8/8), ptosis (4/8), and sensorineural hearing loss (6/8) were common"* (PMID: 41014101).

Genetic / Molecular Information (Section 4)

Causal gene: GAA (acid alpha-glucosidase), 17q25.3, HGNC:4065, OMIM *606800; disease OMIM #232300.

Pathogenic variant spectrum. IOPD arises from two severe/null alleles. In a Mexican cohort, *"Among the 23 different variants identified in the GAA gene, 14 were classified as pathogenic, 5 were likely pathogenic, and 1 was a variant of uncertain significance"* per ACMG/AMP, with **c.-32-13T>G** (the IOPD-associated leaky splice allele) most frequent overall (PMID: 38958145). Representative IOPD nonsense/frameshift variants include **c.1822C>T (p.R608*)**, **c.2662G>T (p.E888*)**, **c.2560C>T (p.R854*)**, and the frameshift **c.236_246delCCACACAGTGC**, the latter homozygous state being *"associated with early disease and a worse prognosis"* (PMID: 34020684).

Population-specific variants. In a Vietnamese cohort, *"The most frequent variants were c.1843G > A and c.1933G > C"* (PMID: 40952111), with two novel variants (c.2016del, c.1723T>C) reported — illustrating founder/population enrichment.

Variant classification & type. Pathogenic/likely-pathogenic per ACMG/AMP; classes span missense, nonsense, frameshift, splice-site, and structural deletions. IOPD genotypes are enriched for null/severe classes; **functional consequence is loss of function** (near-complete enzyme deficiency). All variants are **germline**; somatic origin is not relevant.

CRIM status. Genotype predicts whether any GAA protein is made. Most IOPD patients are CRIM-positive (~85–87%): *"CRIM-positive status was identified in 87.0% of IOPD and 33.3% of LOPD patients"* (PMID: 40952111). CRIM-negative patients (two truly null alleles) make no cross-reacting protein and are at highest immunogenic risk.

Modifier genes / epigenetics / chromosomal abnormalities. No robust modifier genes beyond GAA genotype itself are established; the principal modifier of severity is residual enzyme activity conferred by leaky alleles. No specific epigenetic marks or large-scale chromosomal abnormalities characterize IOPD.

Pseudodeficiency caveat. East Asian pseudodeficiency alleles complicate screening: the c. [752C>T;761C>T] haplotype is *"markedly enriched in East Asian populations (allele frequencies: 0.259% for c.752C > T and 0.263% for c.761C > T)"* and lowers measured enzyme without causing disease (PMID: 42320386).

Environmental Information (Section 5)

IOPD is a purely genetic Mendelian disorder. **No environmental toxins, radiation, pollutants, occupational exposures, lifestyle factors, or infectious agents cause or trigger it.** Intercurrent infection can precipitate cardiorespiratory decompensation in an affected infant (e.g., a 5-month-old presenting with fever and heart failure who was found to carry homozygous c.2662G>T; PMID: 38450370), but such infections are non-specific stressors, not etiologic agents.

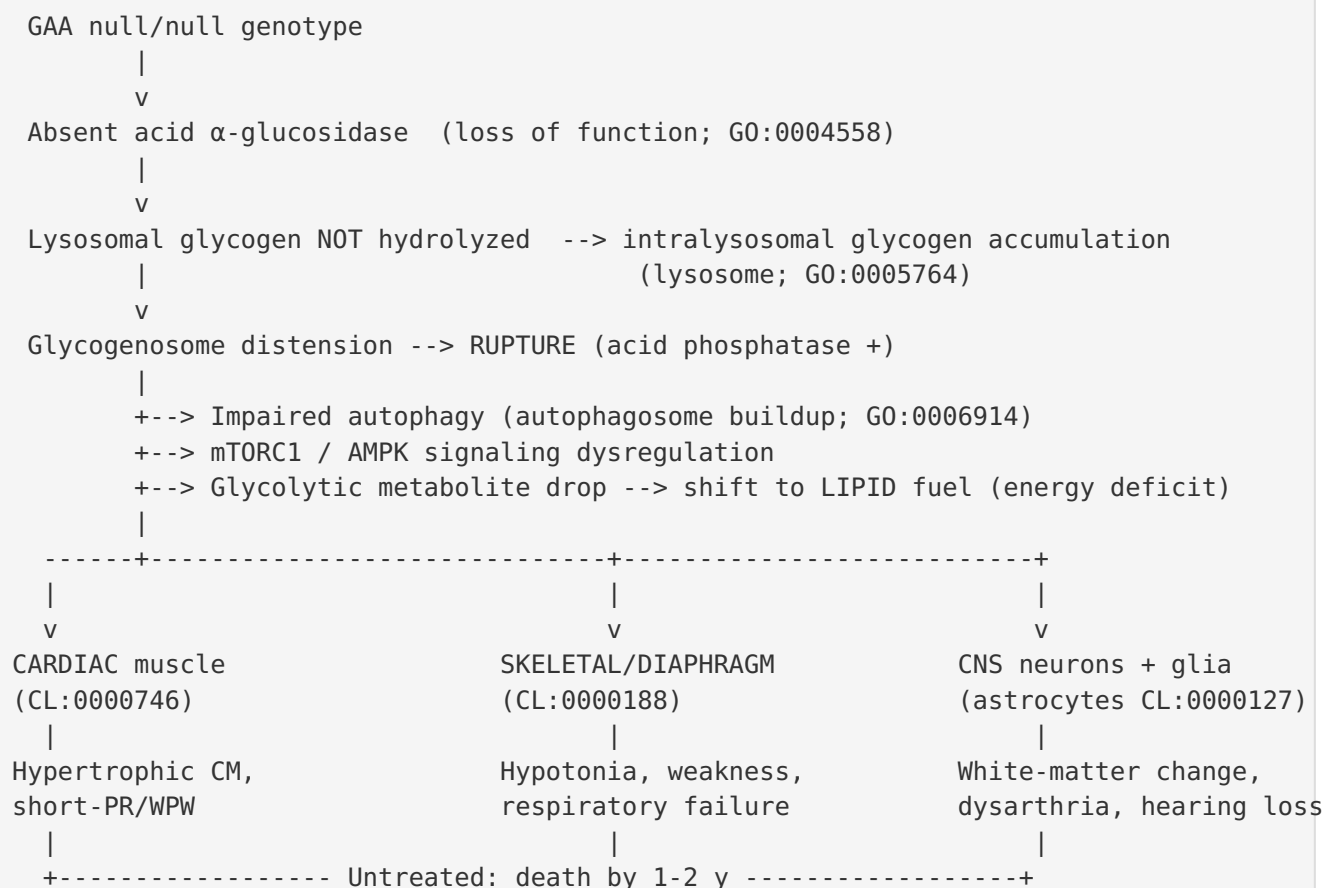
Mechanism / Pathophysiology (Section 6)

Ordered causal chain

1. **Biallelic severe/null GAA variants** → lead to absent or near-absent lysosomal acid α -glucosidase (loss of function). (*Demonstrated.*)
2. **Absent GAA** → results in failure to hydrolyze lysosomal glycogen to glucose → progressive **intralysosomal glycogen accumulation** in cardiac, skeletal, and smooth muscle. (*Demonstrated* — PMID: 40237692.)

3. **Glycogen entrapment** → *leads to* segregation of glycogen during autophagy, phagolysosomal accumulation of undigested glycogen, and eventually **rupture of distended glycogenosomes** (confirmed lysosomal by increased acid phosphatase). (*Demonstrated in canine and human GSD II* — [PMID: 3921759](#).)
4. **Lysosomal dysfunction** → *sets in motion* "extra-lysosomal" events: **defective autophagy** (autophagosome accumulation), **disruption of mTORC1/AMPK signaling**, and a **metabolic/energetic deficit** — decreased glycolytic-pathway metabolites and a **shift to lipids as the energy source** in diseased muscle. (*Demonstrated* — [PMID: 32671132](#); *reversal by gene therapy confirms causality* — [PMID: 37463048](#).)
5. **BRANCH A — Cardiac:** myocardial glycogen storage → *causes* massive **hypertrophic cardiomyopathy** and a conduction signature (short PR, high-voltage QRS, WPW) → *leads to* heart failure/arrhythmia and, untreated, death. (*Demonstrated*.)
6. **BRANCH B — Skeletal/respiratory muscle:** myofiber storage + autophagic buildup → *causes* generalized weakness, hypotonia, diaphragmatic failure → respiratory insufficiency. (*Demonstrated*.)
7. **BRANCH C — CNS (unmasked by ERT-prolonged survival):** neuronal and glial glycogen accumulation (astrocytes, globus pallidus, brainstem motor nuclei, anterior horn cells) → *contributes to* white-matter changes, dysarthria, hearing loss. (*Demonstrated histopathologically* — [PMID: 42538472](#).)
8. **Mitochondrial dysfunction** is reported to accompany lysosomal storage, potentially amplifying cardiac energetic failure. (*Inferred / emerging* — [PMID: 38450370](#).)

ASCII mechanistic model



Molecular / cellular / metabolic detail

- **Molecular pathways:** lysosomal glycogen catabolism (defective); autophagy; mTORC1/AMPK nutrient-sensing signaling (dysregulated) — [PMID: 37463048](#).
- **Cellular processes:** macroautophagy failure, autophagic vacuole accumulation, glycogenosome rupture, cellular energy stress — [PMID: 32671132](#).
- **Protein dysfunction:** loss of function of GAA (misfolding/truncation/absence depending on variant class).
- **Metabolic changes:** decreased glycolytic intermediates, compensatory reliance on lipid oxidation in diseased muscle — [PMID: 32671132](#).
- **Immune involvement:** not autoimmune, but immunogenicity of ERT (anti-rhGAA IgG) is a central therapeutic problem, especially in CRIM-negative patients — [PMID: 39314994](#).

- **Tissue damage mechanisms:** mechanical disruption of myofibers by glycogen-laden/ruptured lysosomes, autophagic buildup, energetic failure.

Suggested GO terms: GO:0004558 (alpha-1,4-glucosidase activity), GO:0005764 (lysosome), GO:0006914 (autophagy), GO:0005977 (glycogen metabolic process), GO:0005980 (glycogen catabolic process). **CL terms:** CL:0000746 (cardiac muscle cell), CL:0000188 (skeletal muscle cell), CL:0000187 (muscle cell), CL:0000127 (astrocyte), CL:0000540 (neuron).

Anatomical Structures Affected (Section 7)

Organ level (primary): heart (myocardium), skeletal muscle, diaphragm, tongue. **Secondary/associated:** liver (hepatomegaly), CNS, and — in canine model — esophagus (dilatation). **Body systems:** cardiovascular, musculoskeletal, respiratory, and (with prolonged survival) nervous system.

Tissue/cell level: striated (cardiac and skeletal) muscle is the principal target; smooth muscle also affected. CNS histopathology shows *"The cerebral and cerebellar white matter (WM), globus pallidus, dentate nucleus, motor nuclei of the brainstem, and anterior horn cells were most frequently affected. Glycogen accumulation was severe in astrocytes and in neurons of the globus pallidus"* ([PMID: 42538472](#)).

Subcellular level: the **lysosome** (GO:0005764) is the primary compartment of pathology; secondary involvement of **autophagosomes** (GO:0005776) and **mitochondria** (GO:0005739).

Localization / lateralization: involvement is **bilateral and generalized/systemic** rather than focal.

Suggested UBERON terms: UBERON:0000948 (heart), UBERON:0001133 (cardiac muscle), UBERON:0001134 (skeletal muscle tissue), UBERON:0001103 (diaphragm), UBERON:0002240 (spinal cord), UBERON:0002037 (cerebellum), UBERON:0001873 (globus pallidus), UBERON:0002107 (liver), UBERON:0001723 (tongue).

Temporal Development (Section 8)

Onset: congenital/neonatal to early infancy; symptom onset typically within the first weeks to 3 months of life ([PMID: 26310554](#)). Onset pattern is **subacute and rapidly progressive**.

Progression: untreated classic IOPD is **relentlessly progressive** with no plateau, reaching end-stage cardiorespiratory failure and death within the first 1–2 years (PMID: 39314994). With ERT, cardiomyopathy regresses and survival is prolonged, converting an acute-fatal course into a chronic, lifelong disease with residual skeletal-muscle and emerging CNS morbidity.

Critical period / therapeutic window: the **neonatal period** is the window of opportunity. Early ERT (≤ 4 weeks, newborn-screening-enabled) improves outcomes: *"These data highlight the benefits of early ERT initiation and ITI, along with high-dose ERT. Despite early treatment, patients with IOPD remain at risk of developing HSAT"* (PMID: 41536902).

Remission: no spontaneous remission. Cardiac hypertrophy is treatment-reversible; skeletal and CNS disease are only partially controlled.

Inheritance and Population (Section 9)

Inheritance: autosomal recessive.

Epidemiology. Birth prevalence of IOPD, from newborn-screening programs, *"ranged from 1 in 297,387 in Japan to 1 in 62,186 in Taiwan"* (PMID: 40329343). In a US program, *"Two newborns were diagnosed with Infantile Onset Pompe Disease (IOPD), and 31 newborns were confirmed to have Late Onset Pompe Disease (LOPD). The incidence of IOPD + LOPD was 1:16,095"* (PMID: 33202836).

Population	IOPD birth prevalence	Source
Japan	1 : 297,387	PMID: 40329343
Taiwan	1 : 62,186	PMID: 40329343
Pennsylvania, USA	IOPD+LOPD combined 1 : 16,095	PMID: 33202836

Penetrance / expressivity: essentially complete penetrance given a severe biallelic genotype; expressivity is relatively consistent within IOPD (uniformly severe), with residual-activity alleles shifting toward LOPD.

Founder effects / population variants: population-specific enrichment exists (e.g., c.1843G>A and c.1933G>C in Vietnamese patients — PMID: 40952111; East Asian pseudodeficiency haplotype — PMID: 42320386).

Consanguinity increases risk of homozygous severe genotypes. **Sex ratio** ~1:1 (autosomal). **Genetic anticipation and germline mosaicism** are not features. **CRIM distribution:** ~85–87% CRIM-positive (PMID: 40952111).

Diagnostics (Section 10)

Enzyme assay: decreased/absent GAA activity is the biochemical hallmark; classic IOPD typically shows <1% of normal activity (PMID: 16860134). Measured by tandem mass spectrometry on dried blood spots (screening) with confirmatory assay.

Biomarker — urinary glucose tetrasaccharide (Glc4/Hex4). The tetraglucose oligomer Glcα1-6Glcα1-4Glcα1-4Glc (Glc4) is diagnostic and treatment-monitoring. Baseline urinary Glc4 and plasma Hex4 are markedly elevated versus controls, and *"Both urinary Glc4 and plasma Hex4 levels decreased after initiation of ERT for all patients. In the four patients with the best overall clinical response in both skeletal and cardiac muscle, levels decreased to within, or near, normal levels during the first year of treatment. In contrast, levels fluctuated and were persistently elevated above the control ranges in those patients with a less favorable clinical response"* (PMID: 15886040). Measured by HPLC-UV and stable-isotope-dilution ESI-MS/MS. **Analyte:** urinary hexose tetrasaccharide (LOINC-type).

Imaging / functional: echocardiography (LV hypertrophy, cardiomegaly), chest radiograph (increased cardiothoracic ratio), and cardiac evaluation (PMID: 18661169).

Electrophysiology (ECG): characteristic **short PR interval, giant high-voltage QRS, and WPW/pre-excitation.** *"ECG showed a shortened PR interval, with huge QRS complexes and biventricular hypertrophy"* (PMID: 18995995); WPW may be the presenting sign — *"The initial symptoms were hypertrophic cardiomyopathy with Wolf-Parkinson-White syndrome"* (PMID: 12162158).

Genetic testing: GAA sequencing (single-gene, panel, WES, or WGS) confirms diagnosis, defines variant pathogenicity, and — critically — **predicts CRIM status** for treatment planning. There has been a *"major shift from direct CRIM testing using western blotting and mutation analysis to CRIM status prediction based on genetic variant analysis"* (PMID: 41583481). Creatine kinase is typically elevated.

Screening: newborn screening (tandem-MS GAA activity on dried blood spots with second-tier GAA sequencing) detects most cases and enables neonatal ERT (PMID: 40329343, PMID: 33202836). Pseudodeficiency alleles cause false positives requiring confirmatory genotyping (PMID: 42320386).

Differential diagnosis: other causes of infantile hypertrophic cardiomyopathy and floppy infant (other glycogenoses, mitochondrial cardiomyopathies, congenital myopathies, spinal muscular atrophy, endocardial fibroelastosis). The GAA enzyme assay and GAA genotype are distinguishing.

Outcome / Prognosis (Section 11)

Natural history (untreated): uniformly fatal in the first 1–2 years from cardiorespiratory failure (PMID: 39314994).

With ERT — survival transformed. The landmark first multinational IOPD trial (n=8, rhGAA 10 mg/kg weekly) reported that *"the median age at death or treatment withdrawal for all patients was 21.7 months, significantly later than expected for patients who were not treated. ... rhGAA improved ventilator-free survival, cardiomyopathy, growth, and motor function in patients with infantile-onset Pompe disease"* (PMID: 16860134). The larger phase IV ADVANCE study (113 patients, 87 IOPD) found *"Overall survival was 98.1% overall, 97.6% IOPD, 100% LOPD; 92.4% remained invasive ventilator-free (93.4% IOPD, 88.7% LOPD)"* (PMID: 29565424). In a real-life cohort, *"Event incidence was significantly lower in the ERT group (HR = 0.06)"* (PMID: 41453391).

Residual morbidity in long-term survivors. Two decades of ERT experience show *"Cardiac hypertrophy was resolved in all; 2 developed arrhythmias requiring intervention. None of the participants required invasive ventilation. Two participants were ambulatory; 6 used wheelchairs. Flaccid dysarthria (8/8), ptosis (4/8), and sensorineural hearing loss (6/8) were common"* (PMID: 41014101); white-matter hyperintensities are present but stay mild-to-moderate with stable cognition.

Mortality where access/timing is limited remains high: in a Vietnamese cohort, *"Despite ERT administration in 52.9% of patients, overall mortality in the infantile group was 60.8%"* (PMID: 40952111).

Prognostic factors: CRIM status, age at ERT initiation, ERT dose, immune-tolerance induction, and antibody titers. Persistently elevated Glc4/Hex4 predicts poorer skeletal-muscle response (PMID: 15886040).

Treatment (Section 12)

Enzyme replacement therapy (mainstay)

Three ERTs are now approved: *"three agents now approved — alglucosidase alfa, avalglucosidase alfa, and cipaglucosidase alfa plus miglustat"* (PMID: 42538546).

Agent	Class / design	Key evidence	NCIT-type concept
Alglucosidase alfa (rhGAA; FDA 2006)	First-generation recombinant human GAA, IV 20 mg/kg q2w	Prolongs ventilator-free survival, reverses cardiomyopathy (PMID: 16860134, PMID: 29565424)	Alglucosidase Alfa
Avalglucosidase alfa	bis-M6P hexasaccharide-enhanced rhGAA (greater M6PR uptake)	Mini-COMET: motor-function trends favor 40 mg/kg q2w; well tolerated ≥97 wk (PMID: 36542086, PMID: 40449831)	Avalglucosidase Alfa
Cipaglucosidase alfa + miglustat	Second-gen bis-M6P-enriched rhGAA + oral enzyme stabilizer	PROPEL switch: better 6MWD/FVC responses; combined OR 4.05 (PMID: 41650629, PMID: 41769220)	Cipaglucosidase Alfa; Miglustat

First-generation rhGAA has intrinsic limitations — *"such as enzyme inactivation at the near-neutral pH of blood, inefficient target cell uptake, and a necessity for complete lysosomal processing once inside target cells"* (PMID: 41769220) — that the next-generation agents address; in adult trials both *"are at least as efficacious as alglucosidase alfa and possess a comparable safety profile"* (PMID: 40471681). Avalglucosidase alfa's design conjugates *"multiple synthetic bis-M6P-containing hexasaccharides to sialic acids present on the enzyme, thus enhancing M6PR targeting, enzyme uptake, glycogen clearance, and therapeutic outcomes"*

(PMID: 40237692). The switch benefit is quantified in PROPEL: *"Overall, 50.8 % versus 13.3 % of patients experienced improvements in 6MWD (% predicted) and/or FVC with cipa+mig versus alg+pbo"* (PMID: 41650629).

Immune tolerance induction (ITI) — essential for CRIM-negative patients

"The inability of CRIM-negative IOPD patients to produce native GAA exposes them to a high risk of development of anti-rhGAA IgG antibody titers, leading to treatment failure" (PMID: 39314994). Prophylactic ITI (rituximab + methotrexate ± IVIG) controls anti-drug antibodies and improves outcomes; ITI benefit also extends to some CRIM-positive patients (PMID: 41583481). Hypersensitivity reactions can be managed by structured desensitization even in very young infants (PMID: 41549377).

Cardiac effect of ERT

ERT rapidly reverses the cardiac phenotype: *"The QRS voltage (SV1+RV6) decreased from 13 to 2.9 mV after 32 weeks of ERT"* (PMID: 18995995); real-world data show *"Hypertrophic cardiomyopathy normalized in all patients, though electrocardiogram abnormalities persisted in 36%"* (PMID: 41576647), and *"most of the treated patients had rapid regression of ventricular hypertrophy in response to ERT"* (PMID: 18661169).

Investigational — gene therapy

Systemic AAV gene therapy *"reversed glycogen storage in all key therapeutic targets — skeletal and cardiac muscles, the diaphragm, and the central nervous system — in both young and severely affected old Gaa-knockout mice"* and reversed autophagy/mTORC1–AMPK abnormalities (PMID: 37463048). However, a human safety signal exists: *"a 49-year-old woman with Pompe disease who developed a sensory neuronopathy after receiving AT845, an investigational adeno-associated virus (AAV) gene replacement therapy"* (PMID: 42142433) — consistent with AAV-associated dorsal-root-ganglion toxicity.

Supportive / rehabilitative

Respiratory support, nutritional/feeding support, physical/occupational/speech therapy, and cardiology management (including arrhythmia care) are integral. Personalized medicine is **genotype- and CRIM-guided**: variant analysis directs ITI decisions and treatment intensity.

Prevention (Section 13)

- **Primary prevention:** none possible for a Mendelian genetic disease; the relevant intervention is **reproductive genetic counseling** for at-risk families (carrier testing, prenatal diagnosis, preimplantation genetic testing). One CRIM-negative case was prenatally diagnosed enabling immediate management ([PMID: 33013846](#)).
 - **Secondary prevention: newborn screening** is the cornerstone — early detection enables pre-symptomatic/neonatal ERT (the critical window) ([PMID: 33202836](#), [PMID: 41536902](#)). Cascade/carrier screening in affected families.
 - **Tertiary prevention:** ITI to prevent antibody-mediated treatment failure; cardiac surveillance for arrhythmia; respiratory and nutritional management to prevent complications.
 - **Immunization / public-health / environmental measures:** not applicable to disease causation (standard infant vaccination remains appropriate given respiratory vulnerability).
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Other Species / Natural Disease (Section 14)

Naturally occurring GSD II occurs across species: *"Naturally occurring animal homologs of LSDs have been described in the mouse, rat, dog, cat, guinea pig, emu, quail, goat, cattle, sheep, and pig"* ([PMID: 15133760](#)). The canine (Lapland dog) natural model is especially faithful: *"Canine GSD II closely parallels the infantile form of the human disease, except for the presence of oesophageal dilatation. Generalized glycogen storage particularly affected muscular tissues (skeletal, oesophageal, cardiac and smooth muscle)"* ([PMID: 3921759](#)).

- **Taxonomy / breeds:** dog (*Canis familiaris*, NCBI Taxon 9615; Lapland dog), cat (*Felis catus*, 9685), Japanese quail (*Coturnix japonica*, 93934), cattle (*Bos taurus*, 9913; Brahman, Shorthorn), sheep (*Ovis aries*, 9940).
 - **Orthologous gene:** GAA is conserved (mouse *Gaa*, NCBI Gene 14618).
 - **Comparative biology:** disease mechanism (lysosomal glycogen storage from GAA deficiency) is evolutionarily conserved, underpinning the translational value of animal models. Esophageal/megaesophagus involvement is a distinguishing canine feature.
 - **Transmission:** none — non-infectious, inherited; no zoonotic potential.
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Model Organisms (Section 15)

Mouse (workhorse model). *Gaa*-knockout mice are the standard model. A modern CRISPR compound-heterozygous knock-in — "*CRISPR/Cas9-mediated knock-in introduced R608 (exon 14) and K889 (exon 18) mutations into C57BL/6 zygotes*" — recapitulates reduced GAA activity, elevated plasma glycogen, cardiac hypertrophy, and diaphragm/skeletal-muscle pathology, more faithfully modeling patient compound-heterozygous genotypes ([PMID: 41742217](#)).

Model	Type	Recapitulation	Key use	Ref
<i>Gaa</i> -KO mouse	Mammalian knockout	Multi-tissue storage incl. CNS; autophagy/mTORC1–AMPK defects	ERT & AAV gene-therapy testing	PMID: 37463048
<i>Gaa</i> R608/ <i>K889</i> knock-in mouse	Compound-het knock-in	Reduced GAA, ↑ plasma glycogen, cardiac hypertrophy, diaphragm pathology	Genotype-faithful modeling	PMID: 41742217
Lapland dog	Natural mammalian	Closely parallels human infantile disease (+esophageal dilatation)	Comparative pathology, large-animal therapy	PMID: 3921759
Japanese quail	Natural avian	Generalized glycogen storage	Natural LSD homolog	PMID: 15133760

Model applications: dissecting downstream pathogenesis (autophagy, signaling, metabolism), and preclinical testing of ERT and AAV gene therapy — AAV correction of storage and secondary abnormalities in *Gaa*-KO mice ([PMID: 37463048](#)) supported clinical translation. Adjunctive β 2-agonists (clenbuterol, salmeterol) that upregulate CI-MPR enhance muscle enzyme uptake in mouse models ([PMID: 24448824](#), [PMID: 30803275](#)).

Model limitations: murine models may under-represent the full human CNS phenotype and long-term skeletal-muscle refractoriness; immune responses to human GAA differ between species. **Resources:** MGI, IMSR for mouse lines.

Mechanistic Model / Interpretation

The unifying interpretation is that **a single enzymatic lesion cascades into multisystem disease**. Absent GAA blocks the terminal step of lysosomal glycogen catabolism; the lysosome fills, distends, and ruptures, and — crucially — the pathology is *not* confined to the lysosome. Downstream "extra-lysosomal" consequences (autophagic block, mTORC1/AMPK dysregulation, a glycolysis-to-lipid metabolic shift) explain why simply restoring enzyme (ERT) reverses cardiac storage dramatically yet incompletely rescues skeletal muscle: cardiomyocytes take up mannose-6-phosphate-tagged enzyme efficiently, whereas skeletal muscle has low CI-MPR density and heavy autophagic burden that impede enzyme delivery. This mechanistic asymmetry is mirrored clinically (cardiac hypertrophy resolves in all long-term survivors while many remain wheelchair-dependent) and biochemically (Glc4/Hex4 normalizes with good skeletal response but stays elevated with poor response). The CNS, historically "silent" because patients died before neurological disease manifested, is now an emerging frontier unmasked by ERT-prolonged survival — and because intravenous rhGAA does not cross the blood–brain barrier, it is precisely the compartment that motivates CNS-penetrant AAV gene therapy. Finally, **CRIM status is the genetic switch that couples genotype to immunology to outcome**: null/null patients make no tolerogenic self-protein, mount neutralizing antibodies, and fail therapy unless tolerized — the single most actionable prognostic variable after age-at-treatment.

Evidence Base

PMID	Contribution	Evidence type
40237692	Defines GAA-deficiency → lysosomal glycogen chain; avalglucosidase design	Review
39314994	Untreated natural history; CRIM-negative immunogenicity	Human clinical
41742217	Compound-het knock-in mouse model	Model organism
26310554	Cardinal phenotype frequencies (case series)	Human clinical
38958145 / 34020684	Variant spectrum, ACMG classification, genotype-severity	Human clinical
42538472	CNS histopathology distribution	Systematic review
32671132 / 37463048	Extra-lysosomal cascade; AAV reversal	Mechanistic / model
3921759	Canine model, glycogenosome rupture	Comparative pathology
16860134 / 29565424	Pivotal ERT survival/ventilator-free data	Human clinical trials
36542086 / 40449831	Avalglucosidase Mini-COMET efficacy/safety	Clinical trial
41769220 / 41650629 / 40471681	Next-gen ERT rationale & efficacy	Review / trial
15886040	Glc4/Hex4 biomarker	Human clinical
40329343 / 33202836 / 42320386	Epidemiology & pseudodeficiency	NBS / scoping
40952111	CRIM distribution, genotype-outcome, mortality	Human clinical
18995995 / 18661169 / 41576647 / 12162158	Cardiac ECG signature & ERT reversal	Human clinical
41536902 / 41453391 / 41014101	Early ERT, real-life events, 2-decade outcomes	Human clinical
42142433	AAV gene-therapy DRG toxicity signal	Human clinical

PMID	Contribution	Evidence type
15133760	Natural LSD animal homologs	Review

Limitations and Knowledge Gaps

- 1. Skeletal muscle remains incompletely treated.** ERT reverses cardiac but not skeletal disease; the mechanistic basis (low CI-MPR, autophagic block) is understood but not therapeutically solved.
- 2. CNS involvement is under-characterized in vivo.** Histopathology is documented, but the natural history of neurocognitive/white-matter disease in long-surviving ERT-treated patients is still emerging, and current IV ERT does not reach the CNS.
- 3. Small numbers / heterogeneity.** IOPD cohorts are tiny; several key findings rest on case series and single-center studies, and in silico digital-twin modeling has been needed to compensate for underpowered trials.
- 4. Long-term next-generation ERT data in IOPD are limited.** Much comparative efficacy data (avalglucosidase, cipaglucosidase+miglustat) come from LOPD; direct head-to-head IOPD evidence is sparse.
- 5. Gene-therapy safety.** The AT845 DRG neuronopathy signal underscores that AAV vectors carry real human toxicity risk requiring dose/route optimization.
- 6. Ontology precision.** MONDO subtype IDs specific to IOPD (vs. parent GSD II) and complete HPO frequency annotations were assigned from best available mappings and should be curator-verified.

Proposed Follow-up Experiments / Actions

- 1. CNS-directed therapy:** develop and test blood–brain-barrier-penetrant delivery (CNS-tropic AAV serotypes, engineered enzyme) in *Gaa*-KO mice with longitudinal MRI/neurobehavioral endpoints, given the documented neuronal/glial storage.
- 2. Skeletal-muscle enhancement:** clinically evaluate CI-MPR-upregulating adjuncts (β 2-agonists) and autophagy-modulating strategies in combination with next-generation ERT.

3. **Prospective CRIM-stratified natural-history registry** capturing age-at-ERT, ITI regimen, antibody titers, Glc4/Hex4 trajectories, and motor/respiratory/CNS outcomes to define genotype–phenotype–treatment–response rules.
 4. **Biomarker qualification:** formally qualify urinary Glc4/plasma Hex4 as a regulatory surrogate endpoint distinguishing cardiac vs. skeletal response.
 5. **Universal newborn screening + neonatal treatment protocols** with harmonized second-tier genotyping to manage pseudodeficiency false positives and enable treatment in the critical neonatal window.
 6. **Head-to-head IOPD trials** of avalglucosidase alfa vs. cipaglucosidase alfa + miglustat, and prophylactic ITI optimization for CRIM-positive as well as CRIM-negative infants.
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Report compiled from 16 evidence-backed findings across 62 reviewed papers. Evidence types span human clinical cohorts and trials, model-organism and comparative-pathology studies, and mechanistic/in-vitro work, as annotated throughout.

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